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https://github.com/pradeep271/K-Means-Clustering.git

Understanding K-Means Clustering: How the Number of Clusters (K) Impacts Machine Learning Performance

Introduction

One of the most crucial methods in unsupervised machine learning is clustering. In contrast to supervised learning, which uses labelled samples to teach algorithms, clustering looks for hidden patterns in unlabelled data. **K-Means** clustering is one of the most used clustering algorithms due to its great effectiveness for exploratory data analysis, ease of comprehension, and computational efficiency.

Using the Mall Consumer Segmentation dataset, this course explores the use of K-Means clustering for consumer segmentation. The main goal is to comprehend how clustering quality and interpretation are impacted by the selection of K, the number of clusters. Choosing a suitable value is essential for getting significant results because K-Means needs the user to choose the number of clusters in advance. Important machine learning techniques are also demonstrated by the research, such as feature scaling, cluster evaluation using the Elbow Method and Silhouette Analysis, business interpretation of clusters, ethical implications, and accessibility enhancements through colourblind-friendly visualizations.

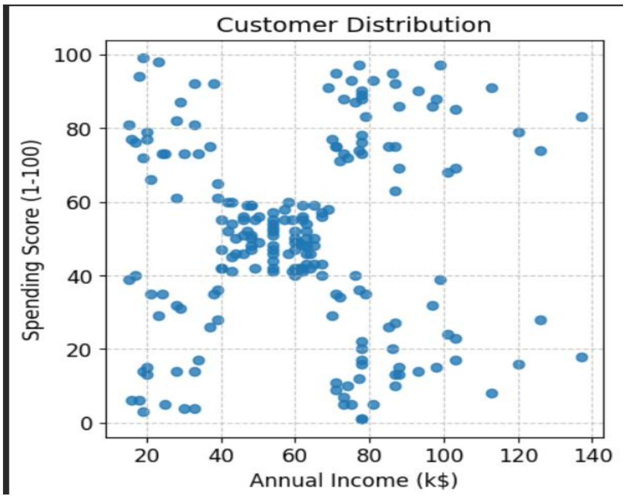
Dataset Exploration

The Mall Customer Segmentation dataset, which was acquired from Kaggle, was utilized in this study. The dataset provides details about mall patrons, including **Customer ID, Gender, Age, Annual Income, and Spending Score**. Annual Income and Spending Score were chosen as the main characteristics for grouping. Because they accurately reflect consumer behaviour and purchasing power, these variables are especially appropriate. Businesses can discover client groups with similar purchasing patterns and develop focused marketing campaigns by analysing these two variables.

Customer points were dispersed over various areas of the feature space, according to preliminary data investigation. The dataset was appropriate for clustering analysis because visual inspection revealed the presence of several natural categories.

Table 1: Dataset Description

Feature Name	Data Type	Description
CustomerID	Integer	Unique identifier assigned to each customer
Gender	Categorical	Gender of the customer
Age	Integer	Age of the customer in years
Annual Income (k$)	Numerical	Annual income of the customer measured in thousands of dollars
Spending Score (1-100)	Numerical	Score assigned by the mall based on customer spending behaviour and purchasing patterns



Data set Exploration

K-Means Clustering

K-Means is an iterative clustering algorithm that partitions data into K distinct groups. The objective is to minimize variation within each cluster while maximizing separation between clusters.

The optimization objective of K-Means is:

$$\arg\min_{\mathbf{C},\mathbf{c}} \sum_{k=1}^{K} \sum_{x \in C_k} \|x - c_k\|^2$$

Within-Cluster Sum of Squares (WCSS) is a popular name for this objective function. To make observations inside each cluster as comparable as feasible, K-Means aims to reduce this value. First, K initial centroids are chosen using the method. Next, using Euclidean distance, each observation is allocated to the closest centroid. New centroids are determined by taking the mean of all locations inside each cluster once all observations have been allocated. Until cluster assignments settle and stop changing dramatically, this process is repeated.

The centroid of a cluster is calculated using:

$$c_k = \frac{1}{|C_k|} \sum_{x \in C_k} x$$

|Ck|: The number of data points currently assigned to cluster

x: The vector sum of all data points in that cluster.

Because they specify the centre of each cluster and have an impact on subsequent allocations, centroids are essential. Centroids advance toward the densest areas of the data with each iteration until convergence is attained. The average consumer attributes within each cluster are represented by the final centroid positions.

Why StandardScaler Was Used

Feature scaling is one of the most important preprocessing steps when applying K-Means clustering. Since K-Means relies on distance calculations, variables with larger numerical ranges can dominate the clustering process and distort results. Therefore, scaling ensures that all features contribute equally to cluster formation.

The Euclidean distance used by K-Means is calculated as:

$$d(x,c) = \sqrt{\sum_{i=1}^{n}(x_i - c_i)^2}$$

xi: The coordinate of the data point in i-th dimension.

ci: The coordinate of the centroid in the i-th dimension.

n: The total number of features (dimensions).

Because distance calculations are sensitive to scale, even moderate differences between feature ranges can influence clustering outcomes. For example, if one feature ranges from 1 to 1000 while another ranges from 1 to 10, the larger feature will dominate distance calculations regardless of its actual importance.

To address this issue, StandardScaler was applied. StandardScaler transforms each feature according to:

where:

$$z = \frac{x - \mu}{\sigma}$$

(x) is the original value

(mu) is the feature mean

(sigma) is the standard deviation

This transformation produces standardized variables with a mean of 0 and a standard deviation of 1. As a result, Annual Income and Spending Score contribute equally to clustering decisions. Standardization improves clustering reliability, enhances convergence, and ensures that the resulting clusters reflect genuine patterns rather than differences in measurement scales.

Elbow Method and the Importance of WCSS

The Elbow Method is one of the most widely used techniques for selecting the number of clusters in K-Means. It is based on the concept of Within-Cluster Sum of Squares (WCSS), which measures how compact clusters are.

WCSS is calculated as:

$$WCSS = \sum_{k=1}^{K} \sum_{x \in C_k} \|x - c_k\|^2$$

K: The total number of clusters.

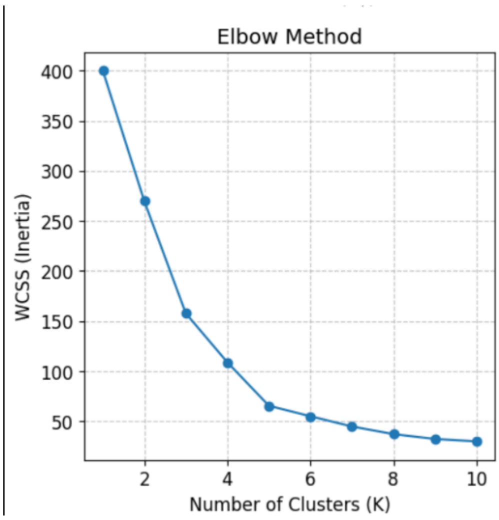
Ck: The set of points in the k-th cluster.

ck: The centroid of the k-th cluster.

||x-ck||^2: The squared Euclidean distance

This metric represents the total squared distance between observations and their assigned centroids. Lower WCSS values indicate that observations are closer to their cluster centres, suggesting more compact clusters. The importance of WCSS lies in its direct relationship with the K-Means objective function. Since K-Means aims to minimize WCSS, monitoring this metric provides insight into clustering quality. As K increases, WCSS naturally decreases because clusters become smaller and observations are closer to centroids. However, continuously increasing K eventually leads to diminishing returns.

The Elbow Method involves plotting WCSS against different values of K. Initially, WCSS decreases rapidly as additional clusters improve cluster compactness. After a certain point, the rate of improvement slows significantly. This point resembles an elbow in the graph and often indicates an appropriate balance between simplicity and accuracy. In this project, the elbow appeared around K = 5, suggesting that five clusters offered a suitable balance between model complexity and clustering performance.



Representation of elbow method

Silhouette Analysis

The Elbow Method is helpful, however determining the elbow location can occasionally be arbitrary. The same graph may be interpreted differently by various experts. Silhouette analysis was used to offer further confirmation. The Silhouette Score compares intra-cluster cohesiveness with inter-cluster separation to determine how well-separated clusters are. A data point's silhouette coefficient can be computed as follows:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

a(i): The average distance between the point i and all other points in the same cluster (measures intra-cluster cohesion).

b(i): The average distance between the point i and all points in the nearest neighbouring cluster (measures inter-cluster separation).

Interpretation:

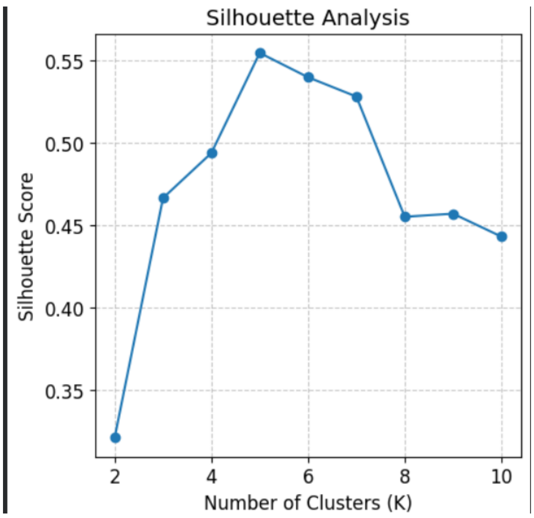
s(i) approx 1: The point is well clustered.

s(i) approx 0: The point lies on or very close to the decision boundary between two clusters.

s(i) approx -1: The point is likely assigned to the wrong cluster.

The range of the silhouette score is -1 to 1. Observations that are poorly matched to nearby clusters and well matched to their own cluster are indicated by values near 1. Negative numbers point to possible misclassification, whereas values close to 0 imply overlapping clusters. The Silhouette Score's significance stems from its capacity to assess cluster quality apart from WCSS. The Silhouette Score takes separation and compactness into account, whereas WCSS concentrates on compactness.

This offers a more thorough evaluation of clustering performance. Silhouette scores were computed for various values of K in this study. The results of the Elbow Method were supported by the greatest scores, which were found around K = 5.



Representation of Silhouette Score

Comparison of different k values

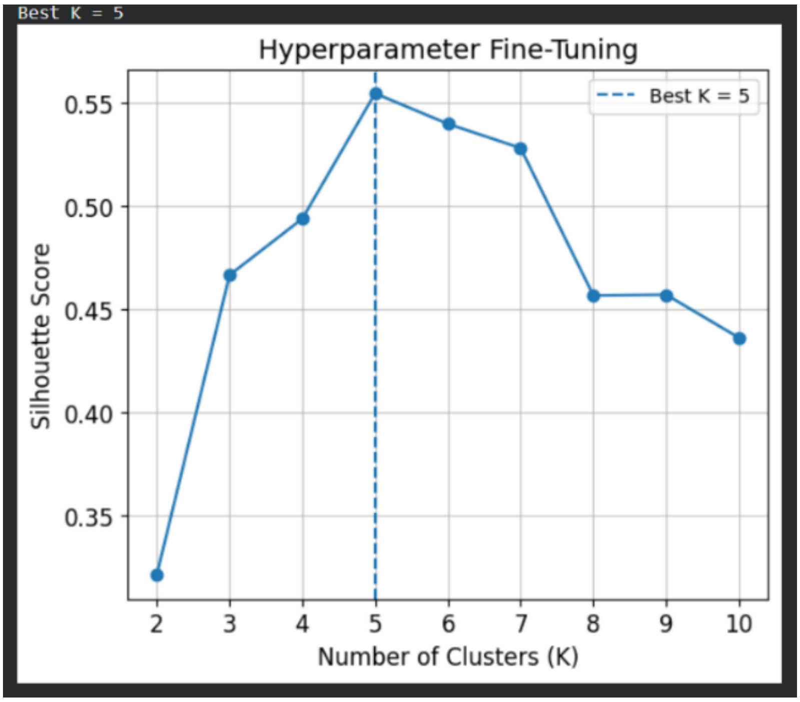
One of the most important decisions when applying K-Means clustering is selecting an appropriate value for K, the number of clusters. To investigate the impact of K on clustering quality and interpretability, clustering results were compared using K = 3, K = 5, and K = 7.

Criteria	K = 3	K = 5	K = 7
Cluster Quality	Moderate	High	Moderate
Cluster Separation	Some overlap between groups	Clear separation between groups	Some clusters become fragmented
Business Interpretability	Broad customer categories	Distinct and meaningful customer segments	More difficult to interpret
Risk	Under-clustering	Balanced clustering	Over-clustering
Marketing Usefulness	Limited targeting opportunities	Strong targeting and segmentation	Reduced practical value

Comparison table and plots

The Importance of Choosing K and Fine-Tuning the Model

Choosing a suitable value for K is one of the most difficult parts of K-Means clustering. K-Means demands that the number of clusters be predetermined, in contrast to supervised learning algorithms where labels direct the learning process. The quality and perception of clustering can be greatly impacted by selecting an inappropriate value. Several natural groups might be combined if K is too small. Under-clustering is the term for these phenomena. Under-clustering makes it harder for the algorithm to identify significant consumer differences and frequently results in segments that are too large and useless. For instance, including all high-income clients in one category might overlook significant variations in their purchasing habits.



Finding Optimal K by hyperparameter tuning

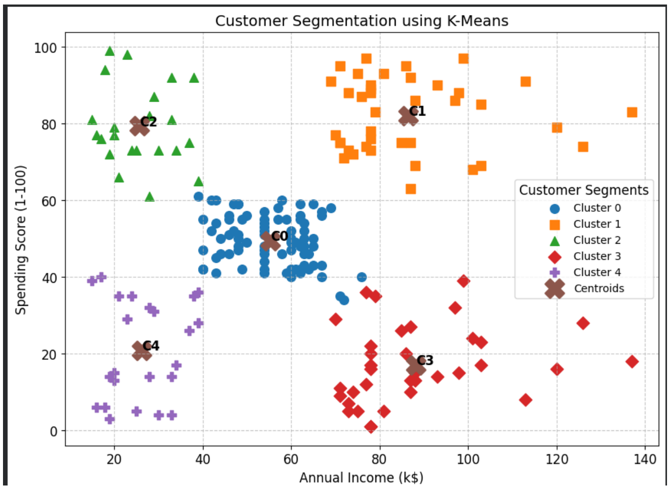
On the other hand, clusters may break apart into smaller artificial groupings if K is too big. We call this over-clustering. Over-clustering may lower WCSS, but it frequently produces segments that are hard to understand and might not offer insightful business information. Additionally, over-clustering can add complexity without adding benefit. Thus, choosing K may be seen of as a way to fine-tune

the model. The practice of modifying model parameters to get optimal performance is known as fine-tuning. The number of clusters is the most crucial factor in K-Means. Results can also be affected by other factors including the value of n_init, the number of iterations, and random initialization. Clusters are made to be both practically helpful and statistically valid by fine-tuning.

Several K values were assessed and contrasted in this experiment. The model was adjusted to determine the best segmentation structure by methodically evaluating various cluster counts and looking at assessment indicators. This procedure highlights how crucial experimentation and validation are to machine learning operations.

Results and Business Interpretation

The final K-Means model was trained with K = 5 after many cluster configurations were assessed. Several separate client groups with distinctive behavioural traits were identified by the segmentation that resulted. Customers with high spending scores and high incomes made up one cluster. These people are high-end clients who make substantial revenue contributions. To improve client retention, businesses should target this market with loyalty programs, special offers, and individualized services. Customers with high incomes but comparatively modest spending habits were found in another cluster. Although they might not be completely involved with the company, these clients have significant purchasing power



Cluster	Characteristics	Business Interpretation
Cluster 1	High Income, High Spending	VIP Customers
Cluster 3	High Income, Low Spending	Potential Customers
Cluster 0	Average Income, Average Spending	Regular Customers
Cluster 2	Low Income, High Spending	Frequent Bargain Shoppers
Cluster 4	Low Income, Low Spending	Low Priority Customers

Customer Segmentation Using K-Means Clustering and keeping K =5

Conclusion, Accessibility, Ethical Considerations, and Limitations

This study uses the Mall consumer Segmentation dataset to show how K-Means clustering can be used for consumer segmentation. Understanding how the number of clusters (K) affects clustering quality and business interpretation was the main goal of the investigation. The study determined that K = 5 was the best value by using StandardScaler, the Elbow Method, and Silhouette Analysis.

This resulted in distinct and significant client categories that may aid in focused marketing campaigns and corporate decision-making. Throughout the project, accessibility was taken into account to guarantee that a variety of customers could comprehend the visuals. To enhance readability, colourblind-friendly marker forms, labelled centroids, high-contrast gridlines, and unambiguous axis labels were employed.

Although K-Means is a simple and effective clustering technique, it has several limitations. The algorithm requires the number of clusters to be specified in advance and may perform poorly when clusters have irregular shapes or contain outliers. Ethical considerations are also important when applying customer segmentation, as clustering may unintentionally reinforce biases if used without proper oversight. Therefore, clustering results should be used responsibly, with human judgement, fairness, and data privacy considerations taken into account.